



Time series prediction of COVID-19 transmission in America using LSTM and XGBoost algorithms

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ABSTRACT

In this paper, we establish daily confirmed infected cases prediction models for the time series data of America by applying both the long short-term memory (LSTM) and extreme gradient boosting (XGBoost) algorithms, and employ four performance parameters as MAE, MSE, RMSE, and MAPE to evaluate the effect of model fitting. LSTM is applied to reliably estimate accuracy due to the long-term attribute and diversity of COVID-19 epidemic data. Using XGBoost model, we conduct a sensitivity analysis to determine the robustness of predictive model to parameter features. Our results reveal that achieving a reduction in the contact rate between susceptible and infected individuals by isolated the uninfected individuals, can effectively reduce the number of daily confirmed cases. By combining the restrictive social distancing and contact tracing, the elimination of ongoing COVID-19 pandemic is possible. Our predictions are based on real time series data with reasonable assumptions, whereas the accurate course of epidemic heavily depends on how and when quarantine, isolation and precautionary measures are enforced.

Introduction

The ongoing outbreak respiratory disease COVID-19 is caused by the novel coronavirus SARS-CoV-2 which happened at the end of 2019 till nowadays, has spread out all over the world and puts tremendous pressure on the economy and society. It was declared as global pandemic by World Health Organization (WHO) on March 11th 2020. Various emergency measures, such as regional lockdown, mass testing, issuing masks, have been taken by many countries to reduce the transmission and control the epidemic. The crucial problems are whether investments in medical services and prevention steps taken are effective in managing the spread of disease, and how the number of confirmed cases will grow in the future.

As the number of cases increases and more data becomes available, various researches [1–7] develop a range of mathematical models or employ machine learning algorithms to forecast the transmission of SARS-CoV-2. Previous studies have also employed LSTM [8–12] or XGBoost [13–19] models to forecast the spread of COVID-19 and identify the most influential COVID-19 indicators. Chimmua et al. [8] adopted LSTM algorithm to forecast confirmed cases in Canada within next two weeks and emphasized the significant role of social distance regular. Tomar and Gupta [9] utilized LSTM and curve fitting to forecast

the number of COVID-19 cases in India for the next 30 days, as well as the influence of preventative measures such as social isolation and lockdown on the spread of COVID-19. Wang et al. [10] focused on predicting the long-term pandemic pattern of COVID-19 employing LSTM networks and a rolling update mechanism. In [11], the implementation of LSTM layers following the proposed convolutional neural network (CNN) block improves the 4-score disease severity prediction performance. Gautam [12] applied transfer learning to LSTM network models to anticipate additional COVID cases and fatalities. Models developed on data from early COVID infected nations such as Italy and the United States are used to predict the development of the disease in other nations. The machine learning algorithm XGBoost was employed to build the models to predict the criticality [13], mortality [14,15] and survival [16] in COVID-19 patients. Li et al. [17] constructed an XGBoost-based classification algorithm to distinguish between influenza and COVID-19 patients. To quantify the impact of the COVID-19 pandemic on driving behavior, the authors [18] utilized explanatory XGBoost feature importance to evaluate the influence of COVID-19 and used seasonal ARIMA models to model. Kukar et al. [19] used random forest, deep neural networks, and XGBoost algorithms to build models that predicted COVID-19 diagnosis based on regular blood test results, age, and gender. To the best of our knowledge, few studies have used the

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